

Signal-Aware Framework for Multilingual Language Model Evaluation

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Motivation

Training mLMs is expensive and requires many decisions at early stages. Practitioners evaluate on large suites, but many benchmarks are noisy or redundant.

Which benchmarks provide **early reliable signal**?

What we did

PRETRAIN 186 models

- 6 sizes (90M-1.7B)
- 11 lang config (1-100L)
- 2 architectures
- 3 seeds

EVALUATE 1.5M data points

- 20 ckpts per model
- BPB on 100 langs.
- 400+ multilingual benchmarks

Research Questions

1. Scaling: What scales predictably?
How: Fit benchmark scaling curves per target language.
2. Prediction: **How early** can we make an informed decision?
How: Compare rankings across model ckpts and sizes.
Finding: BPB predicts ranking after 20% training.
3. Surrogates: Is predictability correlated with **cheaper** metrics?
How: Calculate correlation between metrics and DA.
4. Interventions: Does predictability depend on model-development choices?
How: Compare scaling curves across data mix., archs., etc.
5. Transfer: Does it **generalize** to unmeasured languages?
Finding: Cross-lingual transferability of scaling laws.

- **RQ1 — Scaling: *What scales predictably?***
 - Which multilingual benchmarks follow predictable scaling laws, and what distinguishes benchmarks that scale reliably from those that do not?
 - HOW: Fit scaling curves per benchmark × evaluation language across model sizes/checkpoints. (Compute R^2 , correlation, noise etc.)
 - Expected finding: Identify benchmarks/languages with clean predictable scaling.
- **RQ2 — Prediction: *How early can we make the right decision, and using what benchmark?***
 - How early in model scale or training can we reliably predict the performance and ranking of larger models, and which benchmarks enable the earliest reliable predictions?
 - HOW: At every model size/checkpoint, use benchmark scores to rank the different interventions (data mixture, aspect ratio, etc)
 - Expected finding: Find how much model scale or training compute is needed before the benchmark gives reliable information about the final decision.
- **RQ3 — Surrogates: *Can we know cheaply whether that decision will be reliable?***
 - Which metrics available at small scale are the best surrogates for eventual Decision Accuracy?
 - HOW: Compute candidate metric and measure how well each predicts DA at larger scales.
 - Expected finding: Identify a metric or small set of metrics can be a proxy of DA.
- **RQ4 — Interventions: *Does this remain true when we change model-development choices?***
 - Are scaling behavior and early decision reliability stable across model-development choices, or do interventions fundamentally change which benchmarks and languages are predictive?
 - HOW: Compare scaling curves and decision accuracy across tokenizer, language data mixture, model size, and model aspect ratio etc.
 - Expected finding: Understand if benchmark predictability is an intrinsic property of the benchmark or is conditional on the model/training setup?
- **RQ5 — Transfer: *Does it generalize to languages we haven't measured?***
 - Can scaling behavior and decision-making signals observed in some languages generalize to languages for which we have limited or no scaling observations?
 - HOW: Hold out one evaluation language. Estimate scaling behavior from the remaining languages and predict the held-out language's behavior as scale increases.
 - Expected finding: Quantify cross-lingual transferability of scaling laws.